TAGNet: Temporal Attention Graph Network for Defensive Tactical Pattern Recognition in Professional Counter-Strike 2

Xiang Zhang, RongQi Zhang, Guancheng Chen, Xinlei Cao, ChaoFan Yue, and Chunhong Yuan

***Abstract*—Defensive formations in professional Counter-Strike 2 (CS2) encode multi-agent spatiotemporal coordination, yet how these formations evolve has rarely been modeled. This paper proposes TAGNet, a Temporal Attention Graph Network that recognizes CT-side defensive tactical patterns without tactical labels. TAGNet models the five CT players at five snapshots (10–20 s after the freeze time) as fully connected graphs, encodes each snapshot with a 3-layer distance-aware Graph Attention Network (GAT) trained by self-supervised reconstruction, and aggregates the snapshot embeddings and per-snapshot 25-dimensional context vectors (economy, equipment, utility, score, and previous round) with a Gated Recurrent Unit (GRU) into 64-dimensional round embeddings. On 697 CT defensive rounds from 33 top-tier professional Dust2 matches (HLTV, 2024–2025), TAGNet reaches a test-set reconstruction loss of 0.0086, 8.5% below an earlier two-layer, three-snapshot configuration. Multi-metric K-Means clustering (K=6) discovers six interpretable defensive patterns, from anchor defense (C0) to aggressive forward pressure (C5). CT economy type is the contextual factor most strongly associated with tactical selection (χ^2=96.25, p<0.001, Cramér's V=0.215), and aggressive forward pressure is used significantly more often when trailing (p<0.01). The patterns also yield team-level tactical fingerprints for pre-match scouting. Ablations show that the graph encoder is essential for reconstruction and that the temporal sequence and context input make the patterns more interpretable; six robustness checks, including a domain rule-based baseline and Holm-corrected testing, support these design choices. To our knowledge, this is the first unsupervised temporal graph learning framework for CT-side defensive tactical analysis in professional CS2 on Dust2.**

***Index Terms*—Counter-Strike 2, graph attention networks, temporal graph neural networks, tactical pattern recognition, unsupervised clustering, esports analytics, multi-agent spatiotemporal modeling**

I. Introduction

ESPORTS has evolved into a global competitive industry with growing tournament scale, data density, and tactical complexity [1]. Counter-Strike 2 (CS2) is a round-based tactical shooter in which each round's positioning, resource allocation, and timing encode rich tactical information [2]–[4]. Compared with MOBA games [5], [6] and soccer [7], tactical analysis of first-person shooter (FPS) games is underdeveloped, because FPS data combine high-dimensional states, very short decision windows, and complex multi-agent spatial interactions [8], [9]. Within a short window after the freeze time ends, the five CT players must commit to a positional configuration based on map control, economy, and previous rounds. This setup is closely tied to the round outcome (the CT win rate ranges from 24.2% to 59.1% across the patterns found here), yet no labels exist for such patterns, so they must be discovered automatically. Graph neural networks (GNNs) [10], [11], temporal models [12], [13], graph-temporal fusion in sports [14], and unsupervised learning [15]–[17] make such analysis feasible, yet existing studies mostly target outcome prediction [18]–[21] or macro-level statistics [22], [23].

Despite advances in win/loss prediction [18]–[21], [24], player behavior modeling [22], [25], and competitive data analysis [26], [27], three gaps remain. First, existing CS/CS:GO studies [2], [8] rely on event-level statistics and simple trajectory modeling and do not model the dynamic cooperation between players on a spatial graph. Second, fine-grained modeling of formation evolution within the 10–20-second deployment window is absent. Third, existing methods [3], [8] do not integrate contextual factors such as economy and score pressure into formation embeddings, limiting tactical interpretability. Compared with the win-prediction and macro-statistics literature reviewed in Section II, the novelty of TAGNet lies in its problem formulation and in the pipeline as a whole, which combines established building blocks in a new way. To our knowledge, it is the first pipeline that jointly (i) models CT-side formation evolution as a temporal sequence of fully connected player graphs instead of event logs or single-frame trajectories, (ii) fuses match-context signals into the round embedding before clustering instead of analyzing them post hoc, and (iii) discovers tactical categories without labels and compares them with a domain rule-based positional baseline (Section IV-I). To address these gaps, this paper proposes TAGNet with three contributions:

1) **TAGNet Architecture**: A distance-aware graph attention encoder [10] for the five CT players (22-dimensional node features), in which the pairwise player distance biases every attention score. The encoder is trained without labels by reconstructing each snapshot's

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place distribution, area occupancy, and pairwise distances. The snapshot embeddings and a 25-dimensional match-context vector are aggregated in temporal order by a GRU [12] into a round embedding. The model is lightweight (training takes under 20 s on a laptop CPU), and the final configuration (three attention layers, five snapshots) reduces test-set reconstruction loss by 8.5% relative to an earlier two-layer, three-snapshot configuration (0.0094→0.0086).

2) **CS2 Defensive Tactical Dataset**: Demo files from 33 top-tier HLTV matches on Dust2 (2024–2025), parsed with Awpy [4] into 697 CT defensive rounds, which we release with the trained models and code. Multi-metric K-Means clustering (SC/DBI/CH) discovers six tactically interpretable patterns (C0–C5).
3) **Contextual Factor Analysis**: Chi-square and paired t-tests, Holm-corrected for multiple comparisons, show significant associations of CT economy type (χ^2=96.25, $p_{\mathrm{Holm}}<0.001$, Cramér's V=0.215) and previous-round winner ($p_{\mathrm{Holm}}<0.001$) with tactical selection; aggressive forward pressure (C5) is used significantly more often in deficit states ($p<0.01$). The patterns also yield team-level tactical fingerprints that can support pre-match scouting.

II. Related Work

A. Esports Game Data Analysis and Tactical Mining

Esports analytics has moved from macro-level statistics toward fine-grained tactical understanding. In MOBA games, prior work covers real-time win prediction in Honor of Kings [18], event-level prediction [22], outcome prediction in League of Legends [23], GCN-based semi-supervised win prediction [19], a survey of AI applications [1], and role-sensitive positional analysis [21].

FPS-specific analysis remains scarce: Xenopoulos et al. [2] valued player actions in CS:GO, Xenopoulos and Silva [28] predicted outcomes in American football and Counter-Strike with GNNs on single game-state graphs, Hayakawa et al. [3] predicted VALORANT round outcomes from tactical features extracted from video, and Thai and Scirea [8] modeled CS trajectories with an LSTM. Related sports and game analytics include DyWIS for badminton [20], real-time MOBA player tracking [5], implicit team coordination in Dota 2 [6], process mining for game analytics [26], and a review of spatio-temporal game analytics [9].

B. Graph Neural Networks and Temporal Graph Learning

GNNs model inter-entity relationships through spectral convolution [29], GCN [11], and attention-based aggregation in GAT [10]. Temporal modeling builds on LSTM [13], GRU [12], and self-attention [30]; Temporal Graph Networks [31] and the survey of Longa et al. [32] cover snapshot-based and continuous-time graph learning. In sports tactics, Raabe et al. [14] proposed graph representations of player trajectories for tactical analysis in invasion games and evaluated them on football data.

C. Unsupervised Tactical Pattern Discovery and Cluster Analysis

Clustering enables pattern discovery without expert annotation. This paper uses K-Means [15] with K-Means++ initialization [33], the SC [17], DBI [34], and CH [35] indices, and t-SNE [16]. In games and sports, Drachen et al. [25] modeled players without supervision using self-organizing maps, Cartas et al. [7] combined unsupervised player classification with sequences of player graphs for soccer action spotting, Ke et al. [24] modeled Dota 2 team fights, and Pioroński and Górecki [27] predicted player performance with LightGBM. Unlike these works, TAGNet injects context vectors (economy, equipment, utility, score, and previous-round information) into the round embedding (Eq. (12)), so that clusters reflect both formation structure and context-dependent decisions. To our knowledge, prior FPS defensive analysis has used neither such contextual modeling nor a graph-temporal representation of formations.

III. Methodology

A. Problem Formulation

This paper formulates the CT-side defensive round in professional CS2 matches as a temporal graph sequence learning problem. Let the defensive phase of round r consist of n temporal snapshots, with the snapshot time set denoted as $\mathcal{T} = \{t_1, t_2, \ldots, t_n\}$; here $n = 5$, corresponding to the 10th, 12th, 15th, 18th, and 20th seconds after the end of the freeze time. At snapshot t_k, the five CT players are modeled as a set of graph nodes $\mathcal{V} = \{v_1, \ldots, v_5\}$, and the spatial relationships among players define an undirected edge set $\mathcal{E}$, forming a weighted undirected graph:

$$\mathcal{G}(t_k) = \big(\mathcal{V},\ \mathcal{E},\ \mathbf{X}\big), \quad \mathbf{X} \in \mathbb{R}^{5\times 22} \tag{1}$$

where $\mathbf{X}$ is the node feature matrix and each row is the 22-dimensional state vector of one player (see Section III-C for details). Edge weights are computed by normalized Euclidean distance:

$$w_{ij} = \frac{\|\mathbf{pos}_i - \mathbf{pos}_j\|_2}{d_{\max}} \tag{2}$$

where $\mathbf{pos}_i$ is the planar (x, y) position of player i and $d_{\max}$ is the largest pairwise distance observed in the training split (4215.9 game units), used as the normalization factor; a larger w_{ij} indicates greater spatial separation between players i and j. The objective of this paper is to learn an encoding function f that maps the multi-snapshot graph sequence together with the per-snapshot context vectors $\mathbf{c}_k$ to a fixed-dimensional round embedding:

$$f:\ \big\{\big(\mathcal{G}(t_1), \mathbf{c}_1\big),\ \ldots,\ \big(\mathcal{G}(t_n), \mathbf{c}_n\big)\big\} \longrightarrow\ \mathbf{z} \in \mathbb{R}^d \tag{3}$$

After unsupervised clustering of all round embeddings $\mathbf{z}$, defensive tactical categories are automatically identified. The overall workflow of TAGNet is illustrated in Fig. 1, and the model architecture is shown in Fig. 2.

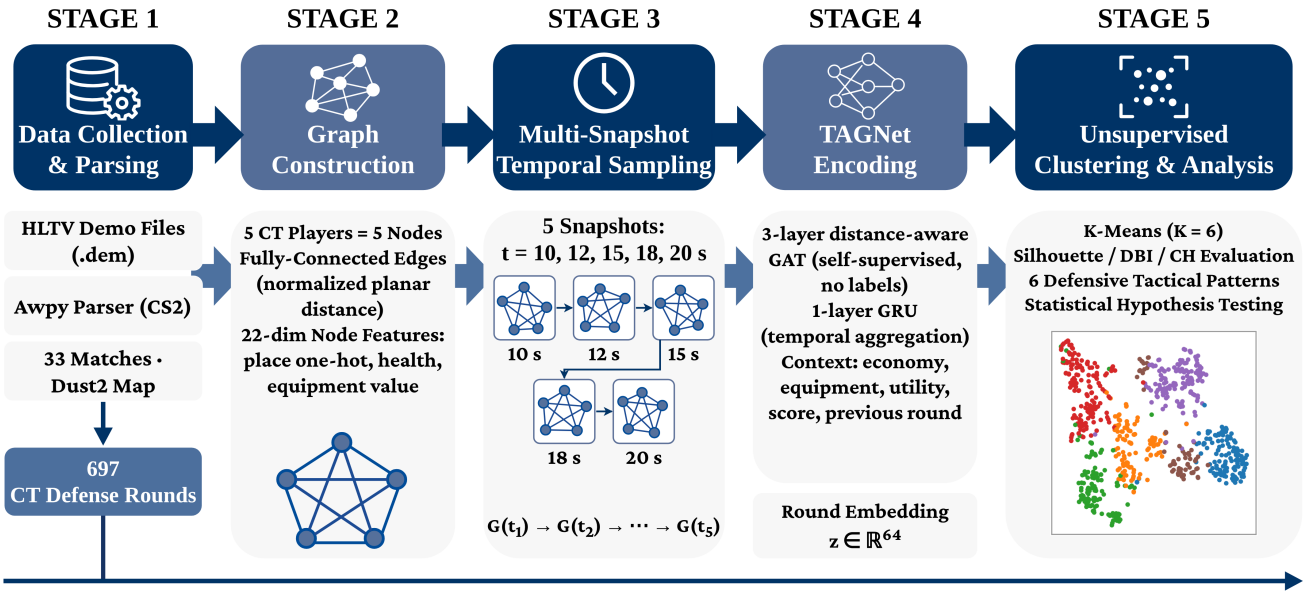


Fig. 1. Overall workflow of TAGNet. The five sequential stages are: data collection and parsing (Stage 1), graph construction (Stage 2), multi-snapshot temporal sampling (Stage 3), TAGNet encoding (Stage 4), and unsupervised clustering and tactical analysis (Stage 5).

TABLE I
DATASET STATISTICS AND PREPROCESSING SUMMARY

Item	Value	Details	Notes
Data source	HLTV.org match replays	Via Awpy	CS2 Demo
Map	Dust2 only	Single-map	de_dust2
Total matches	33	Top-tier	2024–2025
CT rounds retained	697	After filtering	Screened
Train/Val/Test	486 / 95 / 116	≈70:14:16	By match
Graph nodes	17,425	Player nodes	5/snapshot
Graph edges	34,850	Fully connected	Undirected
Context vectors	3,485	Snapshot-level	Econ./score

B. Dataset and Preprocessing

We downloaded the CS2 demo files of 33 top-tier professional matches on the Dust2 map from HLTV and parsed them tick by tick with Awpy [4], extracting player positions, health values, and equipment values. Snapshot ticks are placed at 10, 12, 15, 18, and 20 s after the end of the freeze time (64 ticks per second). To ensure sample quality, the following selection criteria are applied to the CT defensive phase: (a) a round is retained only if five distinct CT players are present in the tick data at every snapshot; players who have already died at a snapshot are kept with zero health (408 of the 17,425 player snapshots); (b) rounds that end before the 20 s snapshot are excluded; (c) rounds with incomplete line-up or snapshot data are excluded. One further match was excluded because its demo header reports a map other than *de_dust2*. A total of 697 CT defensive rounds are retained and split at the match level (23/5/5 matches, approximately 70:14:16 of the rounds) to prevent data leakage. Dataset statistics are summarized in Table I.

C. Graph Construction and Node Features

For each temporal snapshot t_k, the five CT players are modeled as graph nodes, with the node feature vector defined as:

$$\mathbf{x}_i = \left[\,\mathbf{place};\ \mathrm{hp};\ \mathrm{equip}\,\right] \in \mathbb{R}^{22} \tag{4}$$

The semantics of each dimension are listed in Table II, covering a 20-dimensional one-hot encoding of map area (*place one-hot*), one dimension for normalized health value, and one dimension for normalized equipment value, totaling 22 dimensions. The five nodes are fully connected, with edge weights computed by Eq. (2). Full connectivity lets every player attend to all teammates, so the encoder sees the whole formation rather than only local neighborhoods.

In this design, edge geometry and node semantics are kept separate: edge weights are computed from continuous player positions (Eq. (2)), while node features discretize position into named map areas. We tested the alternative of replacing the 20-dimensional *place* one-hot with continuous, train-set-standardized (x, y) coordinates while keeping edges unchanged; as reported in Section IV-I (Table VIII), this raises test-set reconstruction loss 4.2-fold and reduces cross-seed clustering agreement, so the named-area encoding gives higher reconstruction fidelity and stability than raw coordinates.

D. TAGNet Model Architecture

TAGNet consists of three components: a distance-aware graph attention encoder, a temporal fusion module, and a context fusion layer. The architecture is shown in Fig. 2.

1) Distance-Aware Graph Attention Encoder (GAT Module): A 3-layer graph attention encoder with single-head attention and feature dimension 64 is applied to each snapshot $\mathcal{G}(t_k)$, with weights shared across snapshots. In layer l, the

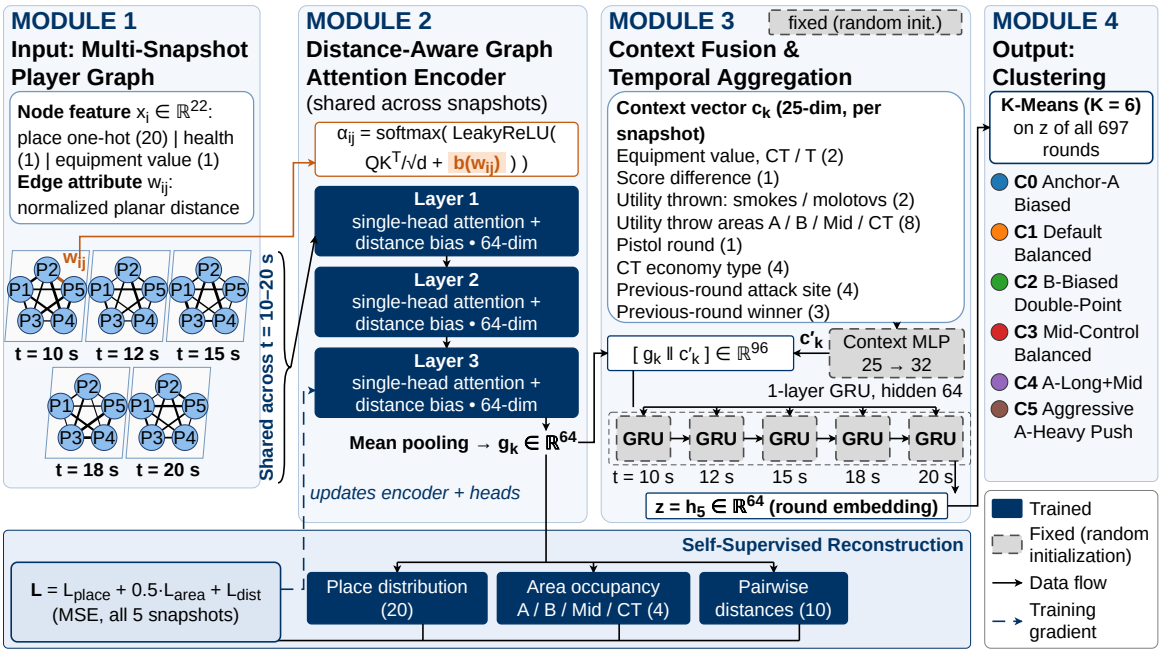


Fig. 2. TAGNet model architecture. MODULE 1: five CT players per snapshot as a fully connected graph (22-dim. node features: 20-dim. place one-hot, health, equipment value; edge attribute: normalized planar distance w_{ij}). MODULE 2: 3-layer single-head graph attention encoder with a learned distance bias $b(w_{ij})$ in every attention score, shared across snapshots and mean-pooled to $\mathbf{g}_k \in \mathbb{R}^{64}$; it is trained by three self-supervised heads that reconstruct, from each $\mathbf{g}_k$, the place distribution, the A/B/Mid/CT occupancy, and the 10 pairwise distances. MODULE 3: the 25-dim. context vector of each snapshot is projected to 32 dims., concatenated with $\mathbf{g}_k$, and aggregated in temporal order by a 1-layer GRU (input 96, hidden 64) used as a fixed aggregator; its final state is the round embedding $\mathbf{z} \in \mathbb{R}^{64}$. MODULE 4: K-Means ($K$=6) yields the six defensive patterns.

TABLE II
NODE FEATURE ENGINEERING FOR CT PLAYER GRAPHS

Feature Category	Feature Description	Dim.
Area enc. (place one-hot)	20-dim. one-hot of named map regions (BombsiteA, LongA, BombsiteB, CTSpawn, Middle, …)	20
Health (norm.)	Min-Max normalized health within $[0, 100]$	1
Equip. value (log1p)	$\log(1+x)$ transformed, then Min-Max normalized by training-set bounds	1
Total node dim.	Concatenated player-level feature vector	22

attention logit from node i to node j combines the similarity of the two players' states with their spatial separation:

$$e_{ij}^{(l)} = \frac{\left(\mathbf{W}_Q^{(l)} \mathbf{h}_i^{(l-1)}\right)^\top \mathbf{W}_K^{(l)} \mathbf{h}_j^{(l-1)}}{\sqrt{d}} + b^{(l)}(w_{ij}) \tag{5}$$

$$\alpha_{ij}^{(l)} = \frac{\exp\left(\mathrm{LeakyReLU}\left(e_{ij}^{(l)}\right)\right)}{\sum_{j' \in \mathcal{V}} \exp\left(\mathrm{LeakyReLU}\left(e_{ij'}^{(l)}\right)\right)} \tag{6}$$

where $\mathbf{W}_Q^{(l)}$ and $\mathbf{W}_K^{(l)}$ are the query and key projections of layer l, d=64, and $b^{(l)}(\cdot)$ is a two-layer perceptron ($1 \to 64 \to 1$, GELU) that maps the normalized distance w_{ij} of Eq. (2) to a scalar bias. Because the graph is fully connected, each player attends to all five players, including itself. Each node then aggregates the value projections $\mathbf{W}_V^{(l)} \mathbf{h}_j^{(l-1)}$, followed by layer normalization (LN) and a residual feed-forward block (FFN):

$$\begin{aligned} \tilde{\mathbf{h}}_i^{(l)} &= \mathrm{LN}\Big(\sum_{j \in \mathcal{V}} \alpha_{ij}^{(l)} \mathbf{W}_V^{(l)} \mathbf{h}_j^{(l-1)}\Big), \\ \mathbf{h}_i^{(l)} &= \mathrm{LN}\big(\tilde{\mathbf{h}}_i^{(l)} + \mathrm{FFN}^{(l)}(\tilde{\mathbf{h}}_i^{(l)})\big) \end{aligned} \tag{7}$$

Placing the distance inside the attention logit lets the encoder learn how strongly spatial proximity should shape the interaction between two players, instead of fixing this weighting by hand. After three layers, global mean pooling over the five players gives the snapshot-level graph embedding:

$$\mathbf{g}_k = \mathrm{MeanPool}\Big(\Big\{\mathbf{h}_i^{(L)} \mid i \in \mathcal{V}\Big\}\Big) \in \mathbb{R}^{64} \tag{8}$$

2) Temporal Fusion Module (GRU Module): At each snapshot step k, the 64-dimensional graph embedding $\mathbf{g}_k$ is concatenated with the 32-dimensional context encoding $\mathbf{c}'_k$ of the same snapshot (Section III-D-3) into $\mathbf{u}_k = [\mathbf{g}_k \| \mathbf{c}'_k] \in \mathbb{R}^{96}$, and the sequence $\mathbf{u}_1, \ldots, \mathbf{u}_5$ is processed in temporal order by a single-layer GRU (input dimension 96, hidden dimension 64) through its reset gate $\mathbf{r}_k$ and update gate $\mathbf{z}_k$:

$$\mathbf{r}_k = \sigma(\mathbf{W}_r[\mathbf{h}_{k-1}; \mathbf{u}_k]), \quad \mathbf{z}_k = \sigma(\mathbf{W}_z[\mathbf{h}_{k-1}; \mathbf{u}_k]) \tag{9}$$

$$\tilde{\mathbf{h}}_k = \tanh(\mathbf{W}[\mathbf{r}_k \odot \mathbf{h}_{k-1}; \mathbf{u}_k]) \tag{10}$$

$$\mathbf{h}_k = (1 - \mathbf{z}_k) \odot \mathbf{h}_{k-1} + \mathbf{z}_k \odot \tilde{\mathbf{h}}_k \tag{11}$$

The final GRU hidden state $\mathbf{h}_5$ serves as the round-level embedding and summarizes the ordered evolution of the defensive formation within the 10–20-second window.

TABLE III
MODEL CONFIGURATION COMPARISON: BASELINE VS. TAGNET

Config.	Snapshots	Layers	FFN	Test Loss
Baseline (earlier)	3	2	shared	0.0094
TAGNet (ours)	5	3	per layer	**0.0086** (↓8.5%)

3) Context Fusion Layer: A context vector $\mathbf{c}_k \in \mathbb{R}^{25}$ is computed at every snapshot, so that utility usage up to that moment is included. It comprises the CT- and T-side equipment values (2 dims.), the score differential (1 dim.), the numbers of smoke grenades and Molotovs thrown by the CT side so far (2 dims.), the shares of these smokes and Molotovs thrown from the A, B, Mid, and CT areas (8 dims.), a pistol-round indicator (1 dim.), a one-hot economy type (4 dims.), a one-hot previous-round attack direction (4 dims.), and a one-hot previous-round winner (3 dims.), totaling 25 dimensions. A two-layer perceptron ($25 \to 32 \to 32$, GELU) projects $\mathbf{c}_k$ to $\mathbf{c}'_k \in \mathbb{R}^{32}$, which enters the GRU input at step k (Section III-D-2). The round embedding is therefore

$$\mathbf{z} = \mathbf{h}_5 \in \mathbb{R}^{64} \tag{12}$$

4) Self-Supervised Training Objective: The graph attention encoder is trained without tactical labels. From each snapshot embedding $\mathbf{g}_k$, three lightweight heads reconstruct the distribution of the five players over the 20 named places ($\mathcal{L}_{\text{place}}$), the occupancy ratios of the four macro areas A, B, Mid, and CT ($\mathcal{L}_{\text{area}}$), and the $\binom{5}{2} = 10$ normalized pairwise distances ($\mathcal{L}_{\text{dist}}$):

$$\mathcal{L} = \lambda_p \cdot \mathcal{L}_{\text{place}} + \lambda_a \cdot \mathcal{L}_{\text{area}} + \lambda_d \cdot \mathcal{L}_{\text{dist}} \tag{13}$$

Each term is a mean-squared error averaged over the five snapshots, with $\lambda_p = 1.0$, $\lambda_a = 0.5$, and $\lambda_d = 1.0$. The encoder and the three heads are optimized with AdamW (learning rate 10^{-3}, weight decay 10^{-4}, batch size 64, gradient clipping at 5.0) for at most 40 epochs with early stopping on the validation loss (patience 10).

Because the reconstruction heads read the snapshot embeddings directly, the objective in Eq. (13) does not depend on the GRU or on the context projection. These two modules therefore keep their seeded random initialization and act as a fixed, order-aware recurrent aggregator of the learned snapshot embeddings and the context, similar to the untrained recurrent reservoirs of echo state networks [36], [37]. All representation learning thus takes place in the graph attention encoder. This keeps the model light (training takes under 20 s on a laptop CPU) and lets the same encoder be combined with other aggregators, as examined in Section IV-I2.

Table III compares TAGNet with an earlier configuration of the pipeline, used here as the baseline, which samples three snapshots (10/15/20 s) and stacks two attention layers that share a single feed-forward block. The final configuration reduces the test-set reconstruction loss from 0.0094 to 0.0086, a relative reduction of 8.5%.

E. Clustering Strategy and Evaluation Metrics

After obtaining all round embeddings, K-Means clustering (scikit-learn, 20 initializations, random state 42) is applied for unsupervised analysis, evaluating $K \in \{2, 3, \ldots, 15\}$ in turn. Clustering quality is measured by three internal metrics jointly. The Silhouette Coefficient (SC) measures intra-cluster compactness and inter-cluster separation, with higher values being better:

$$\text{SC} = \frac{1}{N} \sum_i \frac{b_i - a_i}{\max(a_i,\, b_i)} \tag{14}$$

where a_i is the mean distance between sample i and all other samples in the same cluster, and b_i is the mean distance to the nearest different cluster. The Davies–Bouldin Index (DBI) averages, over clusters, the largest ratio of within-cluster scatter to between-centroid distance, with lower values being better:

$$\text{DBI} = \frac{1}{K} \sum_{i=1}^{K} \max_{j \neq i} \frac{s_i + s_j}{d(\boldsymbol{\mu}_i,\, \boldsymbol{\mu}_j)} \tag{15}$$

where s_i is the mean distance of the members of cluster i to its centroid $\boldsymbol{\mu}_i$, and $d(\boldsymbol{\mu}_i, \boldsymbol{\mu}_j)$ is the distance between two centroids. The Caliński–Harabasz Index (CH) measures clustering quality via the ratio of between-cluster variance to within-cluster variance, with higher values being better:

$$\text{CH} = \frac{\sum_k n_k \|\boldsymbol{\mu}_k - \bar{\boldsymbol{\mu}}\|^2 / (K-1)}{\sum_k \sum_{i \in \mathcal{C}_k} \|\mathbf{z}_i - \boldsymbol{\mu}_k\|^2 / (N-K)} \tag{16}$$

where $n_k = |\mathcal{C}_k|$ and $\bar{\boldsymbol{\mu}}$ is the mean of all N round embeddings $\mathbf{z}_i$. Each metric is converted to a dense rank over the 14 candidate values of K, and the three ranks are averaged. $K=6$ achieves the best average rank (4.33), as summarized in Table IV, which lists the candidates up to $K=8$. Therefore, $K=6$ is selected as the final choice, and all subsequent tactical analyses are conducted based on the six clusters.

TABLE IV
CLUSTERING QUALITY EVALUATION FOR DIFFERENT VALUES OF K (* = FINAL SELECTION). MIN SIZE AND MAX SIZE DENOTE THE SMALLEST AND LARGEST CLUSTER ROUND COUNTS UNDER THAT K. AVG.RANK IS THE MEAN OF PER-METRIC DENSE RANKS OVER K=2–15: SC RANKED DESCENDING, DBI ASCENDING, CH DESCENDING. ROWS FOR K=9–15 ARE OMITTED; THEIR AVERAGE RANKS ARE ALL 5.67 OR HIGHER.

K	SC↑	DBI↓	CH↑	Min size	Max size	Avg.rank↓
2	0.2118	1.8006	207.32	340	357	4.67
3	0.1600	1.9633	153.66	184	317	9.67
4	0.1651	1.9885	136.00	146	194	9.67
5	0.1820	1.7904	126.76	104	169	6.33
6*	0.1835	1.7458	116.09	62	151	**4.33**
7	0.1825	1.7484	110.12	58	135	5.33
8	0.1783	1.7094	104.67	56	117	5.00

Four families of metrics are used: (i) *reconstruction fidelity*, via the multi-task loss $\mathcal{L}$ and its components $\mathcal{L}_{\text{place}}$, $\mathcal{L}_{\text{area}}$, $\mathcal{L}_{\text{dist}}$ (Eq. (13)); (ii) *clustering internal validity*, via SC, DBI, and CH (Eqs. (14)–(16)); (iii) *contextual association*, via chi-square tests with Cramér's V and paired t-tests (Sections IV-F–IV-G); and (iv) *cross-baseline and cross-seed agreement*, via the Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) against alternative baselines and across five random seeds (Section IV-I).

Algorithm 1 TAGNet Training and Inference Pipeline

Input: Demo files $\mathcal{D}=\{d_1,\ldots,d_{33}\}$; snapshot times $\mathcal{T}=\{10,12,15,18,20\}$ s; context features $\mathbf{c}_k$; cluster range $K_{\text{range}}=\{2,\ldots,15\}$
Output: Round embeddings $\mathbf{Z}$; cluster assignments $\mathbf{Y}$; optimal K^*

// Phase 1: Data Preparation
Parse $\mathcal{D}$ with Awpy $\rightarrow$ extract tick-level player states
Filter rounds; retain 697 CT rounds, split $486/95/116$
// Phase 2: Graph Construction
for each round r and snapshot $t_k \in \mathcal{T}$ **do**
  Build $\mathcal{G}(t_k)$ with node features $\mathbf{x}_i$ and edge weights w_{ij} [Eqs. (1)–(4)]
end for
// Phase 3: Encoder Training
Initialize $\theta_{\text{GAT}}, \theta_{\text{dec}}$; initialize the fixed aggregator $\theta_{\text{GRU}}, \theta_{\text{proj}}$ (seed 42)
for epoch $= 1$ **to** E **do**
  for each mini-batch $\mathcal{B}$ **do**
    for each snapshot t_k in $\mathcal{B}$ **do**
      Apply 3-layer GAT [Eqs. (5)–(7)] $\rightarrow \mathbf{h}_i^{(L)}$
      $\mathbf{g}_k=\text{MeanPool}(\{\mathbf{h}_i^{(L)}\})$ [Eq. (8)]
      Decode $\mathbf{g}_k \rightarrow (\text{place}, \text{area}, \text{dist})$
    end for
    Compute $\mathcal{L}$ over all snapshots [Eq. (13)]
    Back-propagate; update $\theta_{\text{GAT}}, \theta_{\text{dec}}$ via AdamW
  end for
  Evaluate on validation set; save best checkpoint
end for
// Phase 4: Round Embedding and Clustering
$\mathbf{z}_r=\text{GRU}(\{[\mathbf{g}_k \| \text{Enc}(\mathbf{c}_k)]\})$ for all r [Eq. (12)]
for K in K_{range} **do**
  Run K-Means$(\mathbf{Z}, K) \rightarrow \mathbf{Y}_K$
  Compute $\text{SC}(K)$, $\text{DBI}(K)$, $\text{CH}(K)$ [Eqs. (14)–(16)]
end for
$K^* = \arg\min_K \text{AvgRank}(\text{SC}(K), \text{DBI}(K), \text{CH}(K))$
return $\mathbf{Z}$, $\mathbf{Y}_{K^*}$, K^*

F. Overall Training and Inference Pipeline of TAGNet

Algorithm 1 summarizes the complete pipeline in four phases: data preparation, graph construction, encoder training, and round embedding with clustering. During training, the reconstruction loss (Eq. (13)) is minimized with respect to the graph attention encoder and the reconstruction heads; the GRU and the context projection act as the fixed aggregator described in Section III-D. At inference, $K^*=6$ is selected by the composite rank of the three metrics (Eqs. (14)–(16)), yielding the six defensive pattern labels.

IV. Experimental Results and Analysis

A. Training Convergence and Choice of the Cluster Number K

Both models ran the full 40 epochs without early stopping; about 95% of the validation-loss reduction occurred in the first 25 epochs, and the validation loss never increased, so neither shows signs of overfitting. TAGNet lowers the test-set loss from 0.0094 to 0.0086 (8.5%, Table III).

For the choice of K, Table IV shows that SC has a local maximum (0.1835) and DBI a local minimum (1.7458) at $K=6$; CH decreases monotonically with K, and its decline slows beyond $K=6$. These local optima of SC and DBI agree with the composite rank, which selects $K=6$ (4.33).

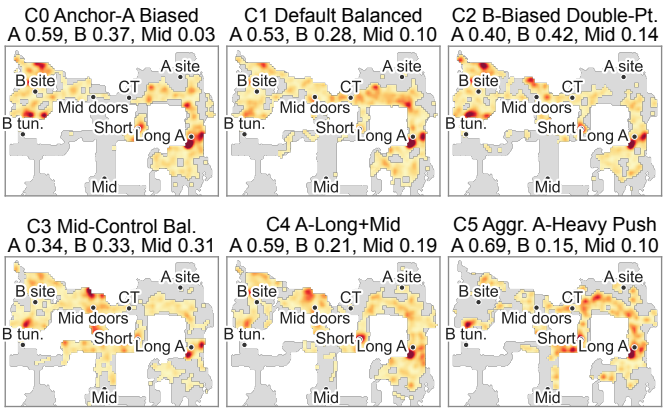


Fig. 3. CT positioning density of the six defensive tactical clusters at $t=15$ s on Dust2. The grey base map is the walkable area estimated from player positions in the parsed demos, and dots mark the median position of each named place in the dataset. Darker red marks positions occupied more often, and all panels share one color scale. The second line of each title gives the share of CT players in the A, B, and Mid areas.

B. Recognition of Six Defensive Tactical Patterns

Fig. 3 shows the CT positioning density of each cluster at 15 s, on which the cluster names are based. **C0 (Anchor-A Biased)**: players concentrate at A and hold a fixed anchor line, with a smaller anchor kept at B. **C1 (Default Balanced)**: players are spread over all regions without a strong concentration, a default setup that is adjusted as information arrives. **C2 (B-Biased Double-Point)**: one high-density point at B and one at A form a two-point defense. **C3 (Mid-Control Balanced)**: Mid occupancy (31.0%) is clearly higher than in the other clusters, so this setup contests the T-side route through Mid. **C4 (A-Long+Mid)**: the defense concentrates on the connected area of A Long and Mid, forming a wide defensive line. **C5 (Aggressive A-Heavy Push)**: the A-site share is the highest of all clusters (68.7%), with players in forward positions such as Long A and Catwalk, which indicates aggressive forward pressure; this is the smallest cluster (62 rounds) and has the lowest CT round win rate (24.2%).

C. Round Embedding Space Visualization

In the t-SNE projection of all 697 round embeddings (Fig. 4), the six clusters occupy largely distinct regions, and training, validation, and test samples are mixed within each cluster, which suggests that the encoder embeds held-out rounds in the same way as training rounds.

D. Temporal Dynamic Evolution of Defensive Formations

Fig. 5 shows that the clusters evolve differently within the 10–20-second window. The A-site ratio of C0 rises only

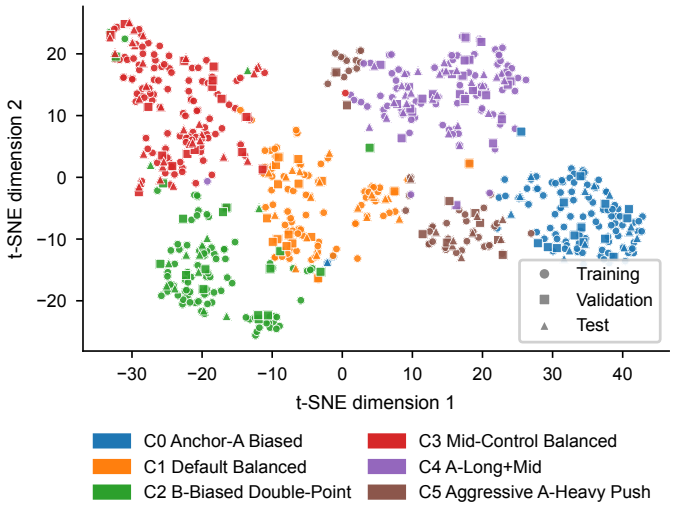


Fig. 4. t-SNE two-dimensional visualization of 697 CT defensive round embeddings. Colors denote the six tactical clusters (C0–C5); shapes (circle/square/triangle) correspond to training, validation, and test sets, respectively.

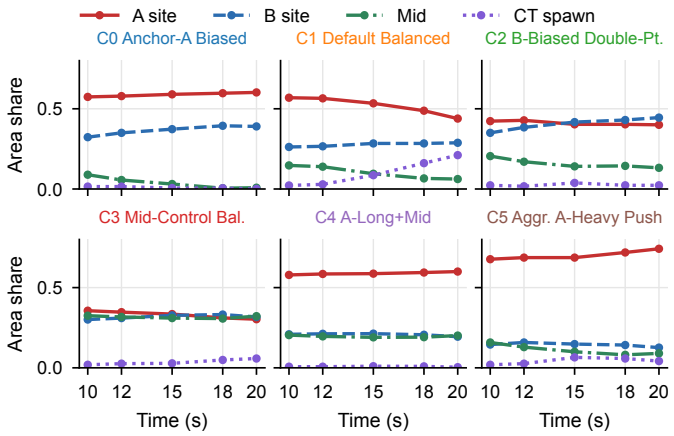


Fig. 5. Area occupancy ratio evolution curves over time (10–20 s) for the six defensive tactical clusters. Red solid lines: A site; blue dashed lines: B site; green dash-dot lines: Mid; purple dotted lines: CT spawn.

slightly (0.57→0.60), as expected for a stable anchor. In C1, the CT-spawn ratio increases after 15 s, which suggests an information-based fallback. In C3, the A, B, and Mid ratios converge at 20 s into a balanced line. C5 has the highest A-site ratio throughout, and this ratio grows over time (0.68→0.74) as the forward pressure on A builds up. None of these dynamics can be seen in a single snapshot, which is why TAGNet models the whole 10–20 s window.

E. Professional Team Tactical Fingerprints and Style Similarity

Fig. 6(a) shows the cluster usage of the 15 teams with the most rounds (out of 26), which forms a distinct tactical fingerprint for each team. *Monte* relies most on C0 (Anchor-A Biased, 0.41), followed by *Astralis* (0.33), consistent with an A-anchor style. *NRG* and *Vitality* both prefer C4 (A-Long+Mid, 0.42 each), and Vitality never uses C2. *Spirit* has the highest C5 share among these teams (Aggressive A-Heavy Push, 0.20).

Fig. 6(b) quantifies style proximity as the cosine similarity of these fingerprints for the 12 teams with the most rounds. *Natus Vincere* and *FURIA* share the highest cosine similarity (0.983), followed closely by *Natus Vincere–Aurora* (0.980) and *FURIA–Aurora* (0.979), so these three teams have very similar tactical distributions. The lowest cosine similarity is between *9z* and *Spirit* (0.625): in Fig. 6(a), *Spirit* never uses C4, which is the most frequent pattern of *9z* (0.29), and relies more on C0 (0.30 vs. 0.13). *Monte*'s similarities span a wide range (0.628–0.898), from *9z* (0.628) to *Falcons* (0.898), which indicates a distinctive tactical style within this dataset.

TABLE V
CHI-SQUARE TEST RESULTS FOR CONTEXTUAL VARIABLES. HOLM p IS THE HOLM-CORRECTED p-VALUE WITHIN THIS 4-TEST FAMILY (SECTION IV-I); SIG. REFLECTS THE HOLM-CORRECTED CONCLUSION.

Contextual Variable	χ^2	Raw p	Holm p	Cramér's V	d.f.	Sig.
CT economy type	96.25	< 0.001	< 0.001	0.215	15	***
Prev.-round winner	26.76	< 0.001	< 0.001	0.201	5	***
Prev.-round attack dir.	27.47	0.025	0.050	0.115	15	n.s.†
Current score state	12.63	0.245	0.245	0.095	10	n.s.

†Significant at the raw threshold ($p<0.05$) but not after Holm correction ($p_{\text{Holm}}=0.050$); treated as suggestive rather than confirmed.

F. Association of Contextual Factors with Tactical Selection

Chi-square independence tests relate four contextual variables to tactical selection (Fig. 7 and Table V).

CT economy type shows the strongest association with tactical selection ($\chi^2=96.25$, $p_{\text{Holm}}<0.001$, $V=0.215$), suggesting that the CT side's buying power is a major factor in its setup. The previous-round winner is also highly significant ($\chi^2=26.76$, $p_{\text{Holm}}<0.001$, $V=0.201$), which may reflect momentum or adjustments made after winning or losing the previous round. The previous-round attack direction reaches significance at the raw threshold ($p=0.025$, $V=0.115$), consistent with the CT side adapting to the T side's recent attacks; however, after Holm correction for this 4-test family (Section IV-I), it no longer reaches significance ($p_{\text{Holm}}=0.050$), and we treat this association as suggestive rather than confirmed. The current score state does not reach significance at either threshold ($p=0.245$), suggesting that short-term score differentials have a limited direct association with specific tactical deployment.

G. Paired Test of Score State on Tactical Usage

To test whether the score state changes tactical usage, paired t-tests compare each cluster's usage proportion between leading and trailing states across teams (Fig. 8). The differences for C0–C4 are not significant, so the use of these five patterns shows no detectable dependence on the score state. C5 (Aggressive A-Heavy Push) is the only cluster with

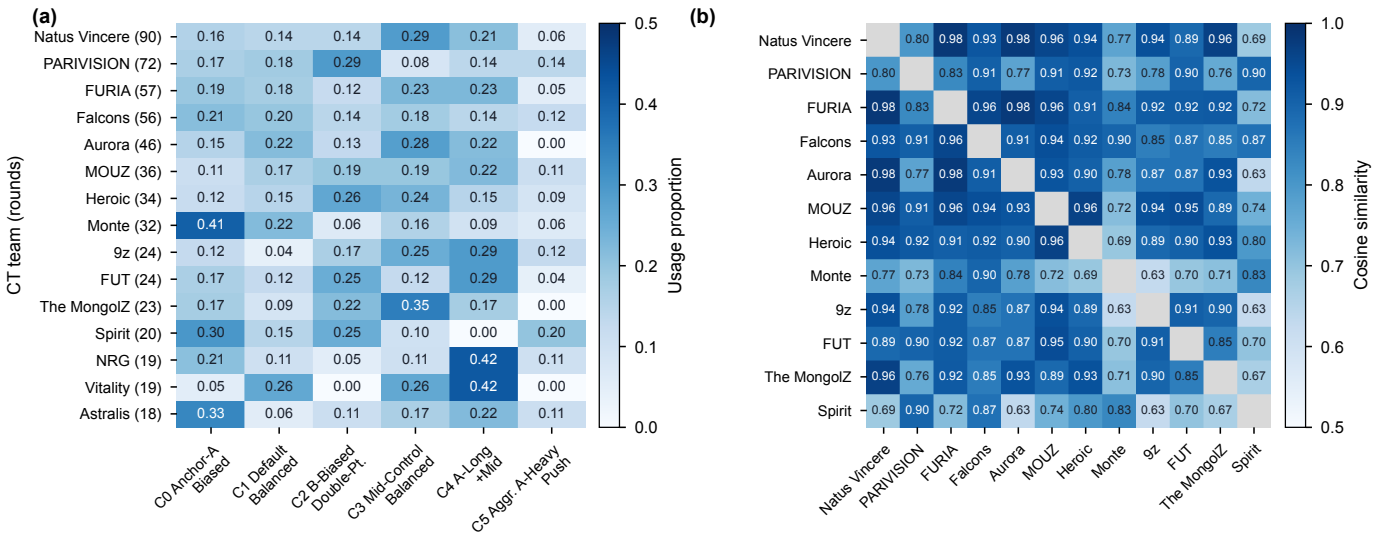


Fig. 6. Professional team tactical fingerprints. (a) Usage proportions of the six defensive tactical clusters for the 15 teams with the most rounds (26 teams in total); darker blue denotes higher reliance. (b) Cosine similarity of the tactical distributions of the 12 teams with the most rounds; three of the teams in (a) are omitted because their round counts are too small for a reliable estimate. Darker blue indicates higher similarity; diagonal cells (self-similarity) are shown in grey.

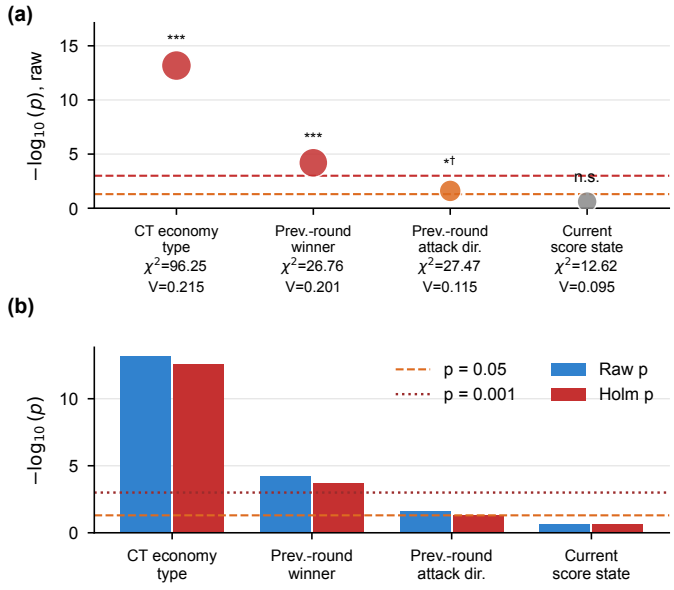


Fig. 7. Chi-square tests of four contextual variables against tactical selection. (a) Raw significance ($-\log_{10} p$) and effect size: bubble size is proportional to Cramér's V, and the χ^2 statistic and V are given under each variable. Colors and stars refer to raw p-values; † marks the association that is not significant after Holm correction. (b) Raw vs. Holm-corrected significance (Table V). Orange dashed and red lines mark p=0.05 and p=0.001.

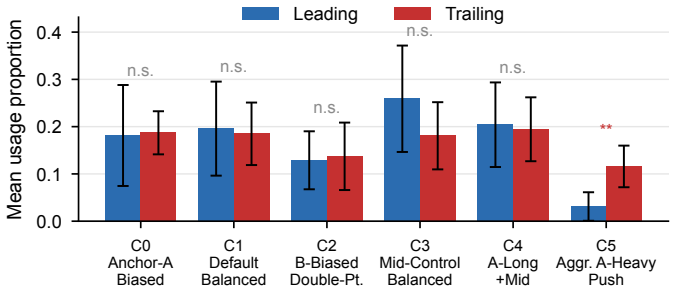


Fig. 8. Comparison of mean usage proportions of six defensive tactical clusters under leading (blue) and trailing (red) score states, averaged over the 22 teams observed in both states. Error bars represent 95% confidence intervals of the team-level mean (t distribution); ** indicates p<0.01 in the paired t-test; n.s. indicates not significant.

a significant difference (p<0.01): its usage proportion in the trailing state (0.116) is higher than in the leading state (0.031), which suggests that trailing teams turn to this high-risk pattern more often in an attempt to recover.

H. Ablation Study

To quantitatively evaluate the contribution of each core module of TAGNet, three ablation variants are designed: removal of context features (*w/o Context Features*), removal of the temporal module (*w/o Temporal Module*), and removal of the graph structure (*w/o Graph Structure*, replaced by an MLP). Table VI compares them with the complete model.

Beyond the four quantitative metrics (test loss, SC, DBI, CH), we report a qualitative *semantic validity* rating, obtained by manually checking whether each discovered cluster maps to a distinct, domain-recognizable tactical pattern (criterion in the caption of Table VI).

As Table VI shows, removing the graph structure raises the test loss sharply to 0.0143 (+66.3%) and collapses the optimal cluster number to K=3, so only very coarse categories remain; its high CH (165.90) is an artifact of this collapse, since three coarse clusters mechanically inflate the between-cluster variance ratio. Modeling the spatial relationships between players is therefore essential for reconstruction. Removing the temporal module (15 s snapshot only) raises the test loss to 0.0090 (+5.2%); its higher silhouette coefficient (0.2074 vs. 0.1788) reflects a more concentrated single-snapshot embedding with superficially sharper boundaries, while the information on how

TABLE VI
ABLATION STUDY RESULTS. "SEMANTIC" DENOTES THE QUALITATIVE INTERPRETABILITY OF DISCOVERED CLUSTER PATTERNS: *Yes* = ALL SIX CLUSTERS YIELD CLEAR TACTICAL LABELS CONSISTENT WITH DOMAIN KNOWLEDGE; *Partial* = MAJORITY OF CLUSTERS ARE INTERPRETABLE BUT ONE OR MORE LACK DISTINCT TACTICAL MEANING; *No* = CLUSTERS REDUCE TO COARSE POSITIONAL GROUPINGS WITHOUT TACTICALLY MEANINGFUL DIFFERENTIATION. SC, DBI, AND CH ARE COMPUTED ON EACH VARIANT'S TRAINING-STAGE K-MEANS PARTITION AT ITS OWN SILHOUETTE-OPTIMAL K (FULL MODEL K=8; W/O CONTEXT AND W/O TEMPORAL K=7; W/O GRAPH K=3), AND ARE THEREFORE NOT DIRECTLY COMPARABLE WITH THE K=6 VALUES IN TABLE IV.

Model Variant	Loss↓	SC↑	DBI↓	CH↑	Semantic
TAGNet (full)	**0.0086**	0.1788	1.7094	104.67	Yes
w/o Context Feat.	**0.0086**	0.1857	1.6716	100.44	Partial
w/o Temporal Mod. (GRU)	0.0090	0.2074	1.6684	108.95	Partial
w/o Graph Structure	0.0143	0.1716	1.7579	165.90	No

the formation evolves within the 10–20-second window is lost. Removing context features leaves the test loss unchanged (0.0086), as expected because the context does not enter the reconstruction loss, but lowers semantic validity to *Partial*. The context input therefore helps the clusters separate context-dependent tactical patterns without affecting reconstruction. Only the complete TAGNet combines the lowest test loss with fully interpretable clusters (semantic validity *Yes*).

I. Supplementary Robustness Checks

Six additional experiments test TAGNet's design choices: a domain rule-based positional baseline, a check of temporal-window necessity, a geodesic edge-distance variant, graph-free temporal baselines, Holm-corrected significance testing, and a node/edge feature-granularity check. Unless noted otherwise, each trained variant is re-trained from scratch under five random seeds ($\{42, 137, 271, 503, 911\}$) on the same 486/95/116 train/val/test split as the main experiments, and we report mean $\pm$ standard deviation across seeds together with the Adjusted Rand Index (ARI) between the seeds' cluster assignments as a stability measure.

1) Domain Rule-Based Positional Baseline: To test whether a manually defined, zero-training-cost positional rule already captures most of what TAGNet discovers, we built the baseline directly from named-area player counts at the final (t=20 s) snapshot: each round is assigned to one of {*balanced*, *A-leaning*, *B-leaning*, *mid-control*, *A-heavy*, *B-heavy*} by a fixed threshold rule on the A/B/Mid player counts, applied in the following order: *mid-control* if at least two players are in Mid; otherwise *A-heavy* if at least four are at A; otherwise *B-heavy* if at least three are at B; otherwise *A-leaning* or *B-leaning* if one site holds at least two more players than the other; otherwise *balanced*. The B-heavy threshold is lower than the A-heavy threshold because default Dust2 CT setups commit fewer players to B than to A: at t=20 s, at most two players are at B in 92.3% of rounds, and at most three are at A in 93.8%. Because *B-heavy* is triggered at three B players, *B-leaning* can only be assigned when no player is at A and exactly two are at B, and this configuration does not occur in the 697 rounds. B-weighted setups are therefore present but fall under *B-heavy* (54 rounds, 7.7%), including 11 rounds with a 1A–3B–1Mid split, the mirror of the common *A-leaning* split 3A–1B–1Mid (127 rounds). The rule also understates milder B-weighted setups: of the 115 rounds in C2 (B-Biased Double-Point), the rule labels 87 *balanced* and 26 *B-heavy*, although 110 of these rounds have at least as many players at B as at A at t=20 s. We report the five realized classes rather than forcing a spurious sixth. We compare this fixed rule against (i) K-Means applied directly to the five-snapshot player-count sequence (a stronger, still training-free positional baseline) and (ii) TAGNet.

TABLE VII
DOMAIN RULE-BASED POSITIONAL BASELINE VS. TAGNET. ARI/NMI ARE AGREEMENT WITH TAGNET'S K=6 PARTITION (SEED 42).

Method	SC↑	DBI↓	CH↑	ARI	NMI
Count-sequence K-Means (K=6)	0.3005	1.2909	204.58	0.375	0.440
Fixed count→location rule (5 classes)	0.1663	1.6606	123.67	0.301	0.446
TAGNet (seed 42)	0.1835	1.7458	116.09	n/a	n/a

As shown in Table VII and Fig. 9(a), the count-sequence K-Means scores better than TAGNet on all three internal clustering metrics, and the fixed rule scores better on DBI and CH. These compactness indices are not directly comparable across feature spaces, and they favor the count features, which take few distinct values (429 of the 697 rounds share their exact five-snapshot count sequence with another round). More informative is the agreement: TAGNet's partition is only moderately related to the player-count configuration (ARI = 0.301–0.375, NMI = 0.440–0.446), so the embedding captures structure beyond the number of players per area, including spacing, health, equipment, and match context. Of the three methods, only TAGNet can therefore separate context-dependent variants of otherwise similar formations. Only 37 of the $\binom{9}{4}$=126 combinatorially possible count configurations across the five area bins (A, B, Mid, CT, other) occur at t=20 s, so the configuration space that occurs in practice is far smaller than the combinatorial bound, which reduces, but does not remove, the concern over dataset size. Collecting data over several seasons in which Dust2 is unchanged is useful future work, provided that changes in team strategies over time are taken into account.

2) Necessity of the Temporal Window: We tested whether the five-snapshot temporal window is necessary by training a single-snapshot variant restricted to t=20 s only, and a variant that mean-pools all five snapshot embeddings instead of using the GRU.

Two results in Table VIII require explanation. First, the Full GAT+GRU and Mean-pool rows report an identical test loss (0.008436) because the reconstruction decoders act on each snapshot's own graph embedding before temporal aggregation; the round embedding is used only for K-Means. The loss is therefore invariant to the aggregator, whereas the clustering metrics, and especially cross-seed ARI, are not. Since neither aggregator is updated by the reconstruction loss

TABLE VIII
SUPPLEMENTARY VARIANTS (MEAN ± STD OVER 5 SEEDS). EACH BLOCK CHANGES ONE COMPONENT OF THE FULL MODEL (FIVE SNAPSHOTS, EUCLIDEAN EDGES, *place* ONE-HOT); FOR GEODESIC + HINGE, THE TEST LOSS INCLUDES THE HINGE TERM.

Check (Sec.)	Variant	Test Loss	SC	DBI	CH	X-seed ARI
Reference	Full GAT+GRU	0.008436±.000138	0.1765±.0082	1.7661±.0737	110.05±8.42	0.5226
Temporal window (IV-I2)	t=20 only (1 snap.)	0.012240±.000307	0.1977±.0155	1.6766±.0862	121.32±11.51	0.5217
	Mean-pool (5 snap.)	0.008436±.000138	0.1967±.0120	1.5899±.0804	144.92±15.09	0.4513
Edge distance (IV-I3)	Navmesh geodesic edges	0.008654±.000322	0.1760±.0069	1.8042±.0419	109.88±5.82	0.5065
	Geodesic + hinge	0.013095±.000499	0.1896±.0121	1.7241±.0187	135.13±10.35	0.5585
Graph-free (IV-I4)	No-graph MLP+GRU	0.014488±.000339	0.1600±.0158	1.7449±.0740	122.23±6.42	0.5442
	No-graph MLP+LSTM	0.014488±.000339	0.1609±.0134	1.7010±.0753	120.63±15.02	0.4936
Node features (IV-I6)	Continuous (x, y)	0.035535±.003907	0.2878±.0196	1.1373±.0641	276.70±28.82	0.3990

(Section III-D), this comparison isolates the aggregation step and is the most direct test of order-aware aggregation: with the same encoder and the same loss, replacing the GRU by a mean over the five snapshots lowers cross-seed agreement from 0.5226 to 0.4513. Second, the 45% higher loss of the t=20-only variant does not isolate the need for history, because the decoders bypass aggregation. It more likely reflects training on one-fifth as many (snapshot, label) examples and on the round's most spread-out snapshot, which is the hardest to reconstruct, whereas the full model averages its loss over all five snapshots. This may also explain why the effect is larger than in the single-snapshot ablation of Table VI, which uses the middle of the window (t=15 s, +5.2%). We therefore treat the t=20-only result as secondary evidence for using the full 10–20 s window and the GRU-vs-mean-pool comparison as the primary evidence for order-aware recurrent aggregation.

3) Geodesic Edge Distance and Consistency Constraint: Euclidean distance can misrepresent in-game reachability (e.g., two adjacent-looking positions separated by a wall). We replaced Eq. (2)'s Euclidean edge weights with Dust2 navmesh geodesic distances between named-place centers, and additionally tested a hinge consistency term (λ=1) that penalizes reconstructed pairwise distances falling outside each area-pair's train-set Euclidean range.

Geodesic edges alone neither improve reconstruction (loss 0.008654 vs. 0.008436) nor consistently improve the clustering metrics. Adding the hinge term (reconstruction-only loss 0.009637±0.000350; mean test-set hinge violation 0.003457) trades reconstruction fidelity for better clustering geometry and higher cross-seed stability (ARI 0.52→0.56). We therefore treat this experiment as a sensitivity check of the assumption that Euclidean distance approximates reachability: geodesic distances and a distance-consistency constraint shift the balance between reconstruction and clustering instead of improving both. Euclidean edges thus remain a sound default for Dust2.

4) Graph-Free Temporal Baselines: To isolate the graph encoder's contribution from the temporal unit's choice, we trained no-graph variants (flattened per-player features through an MLP in place of the GAT) with both GRU and LSTM temporal cells, the latter serving as a lightweight version of the LSTM-based trajectory modeling of Thai and Scirea [8].

As in Section IV-I2, the No-graph MLP+GRU and MLP+LSTM rows share an identical test loss for the same architectural reason: the reconstruction decoders do not depend on the temporal cell. Removing the graph structure raises the test loss by 71.7% with either cell, in line with the +66.3% of the *w/o Graph Structure* ablation in Table VI. The reconstruction gain therefore comes from modeling the interactions between players on the graph, not from the choice of temporal cell; GRU and LSTM differ mainly in the clustering metrics.

TABLE IX
HOLM CORRECTION SUMMARY BY TEST FAMILY.

Test Family	# Tests	# Sig. (Raw)	# Sig. (Holm)
Contextual variables	4	3	2
Score state (paired t)	6	1	1
Team vs. opponent (χ^2)	17	1	0

5) Multiple-Comparison Correction: We applied Holm correction within three pre-declared test families: the four contextual-variable chi-square tests (Table V), the six leading/trailing paired t-tests (Fig. 8), and a 17-test family of per-team chi-square tests of tactical-cluster usage against specific opponents (formalizing the qualitative comparisons in Section IV-E).

The main findings survive the correction: CT economy type and the previous-round winner remain significant (Table V), and C5 (Aggressive A-Heavy Push) remains significant in the score-state family (raw p=0.0020, Holm p=0.0118; Fig. 8). Only the previous-round attack direction loses significance. Within the team–opponent family, the single raw-significant pairing (Aurora, raw p=0.0119) does not survive correction (Holm p=0.2028); we therefore do not claim any statistically robust team-specific opponent effect, and present Fig. 6 as descriptive tactical fingerprints rather than as evidence of significant opponent-conditioned adaptation.

6) Node/Edge Feature Granularity: To address the mismatch between edge weights (computed from continuous positions) and node features (discretized to 20 named areas), we replaced the *place* one-hot with train-set-standardized continuous (x, y) coordinates, keeping health, equipment, and edge weights unchanged.

Continuous coordinates raise the reconstruction loss 4.2-fold and lower cross-seed agreement (ARI 0.52→0.40), although SC, DBI, and CH improve. As with the single-snapshot

ablation in Table VI, better internal metrics here come with lower stability, so they do not by themselves indicate more meaningful clusters. The named-area encoding therefore balances fidelity and stability better, and since the areas are the callouts that players use, it keeps the patterns easy to interpret.

7) Summary of the Robustness Checks: Fig. 9(b) summarizes test loss and cross-seed ARI across all variants in Sections IV-I2–IV-I6. Taken together, Sections IV-I1–IV-I6 show that no single configuration is best on every metric: the count-sequence baseline scores higher on SC/DBI/CH (Table VII), the geodesic-plus-hinge variant on clustering quality and stability, and the continuous-XY variant on SC/DBI/CH (Table VIII). TAGNet, however, attains the lowest test loss of all trained variants (shared with mean-pooling, which uses the same encoder), and no variant matches this loss while also exceeding its cross-seed agreement. Each alternative that improves a clustering metric pays in reconstruction fidelity, stability, or both: continuous XY increases the loss 4.2-fold and lowers ARI, and the hinge term increases the total loss by about 55%. Together with the module-level ablations of Table VI, this makes TAGNet's configuration the best overall trade-off among the variants tested.

V. Discussion

A. Design Rationale and Limitations of TAGNet

Three design choices underpin TAGNet: distance-aware attention; self-supervised reconstruction, which removes the need for tactical labels; and order-aware aggregation. The experiments support the first and the third: removing the graph raises the test loss by 66.3%, and mean-pooling instead of the GRU lowers cross-seed agreement from 0.52 to 0.45. The 8.5% gain over the earlier configuration combines three changes (Table III) and is not attributed to any one of them.

As a first study, this work also has limitations that point to clear next steps. First, TAGNet is trained on a single map (Dust2); transferability to more complex layouts such as Mirage or Inferno is unverified. Maps with strong vertical structure, such as Nuke and Vertigo, pose a further challenge: the edge distance in Eq. (2) uses planar (x, y) positions and the place one-hot does not separate floors, so players on different levels of the same area would appear close. Extending the pipeline to these maps would require three-dimensional or navmesh-based distances (cf. the geodesic variant in Section IV-I3) and level-aware node features. Second, the 22-dimensional node features lack behavioral signals such as movement direction, viewing orientation, and weapon type, which may cause information loss for certain tactical decisions. Third, cluster label assignment (e.g., C5 as aggressive forward pressure) involves interpretation. Draft labels were produced by a fixed threshold rule on each cluster's area occupancy at 15 s. The corresponding author (C. Yuan), who has played CS:GO and CS2 for more than five years (over 5,000 hours) and holds S rank on the Perfect World competitive platform in China, then checked them against the cluster heatmaps (Fig. 3) and representative rounds. Weak supervision or independent expert annotation could make the labels more reliable. Fourth, the round-level assignments vary across training runs (cross-seed ARI ≈ 0.52, Section IV-I); the six-cluster partition analyzed in Sections IV-B–IV-G is the one obtained from the released model. Finally, the GRU and the context projection are used as a fixed aggregator; training them jointly with a round-level objective is a natural extension. Context fusion is nonetheless supported by the ablation, in which removing context lowers semantic validity from *Yes* to *Partial*.

B. Practical Interpretation and Applications in Tactical Intelligence

The six clusters range from conservative (C1 Default Balanced, CT win rate 49.5%) to highly aggressive (C5 Aggressive A-Heavy Push, CT win rate 24.2%), and C2 (B-Biased Double-Point) has the highest CT win rate (59.1%). C3 has the highest Mid occupancy (31.0%), consistent with the common Dust2 mid-control strategy, and C0 and C4 both concentrate on A but differ in that C0 holds a fixed anchor line while C4 links A Long and Mid.

CT economy type is the contextual factor most strongly associated with tactical selection (χ^2=96.25, V=0.215): full-buy rounds most often use C4 (A-Long+Mid, 24.4% of full-buy rounds) and C0 (22.7%), whereas C5 (Aggressive A-Heavy Push) is concentrated in eco rounds (19 of 60 eco rounds, 31.7%, against 14 of 409 full-buy rounds, 3.4%), consistent with riskier setups being chosen when resources are scarce. The higher C5 usage when trailing (p<0.01) points in the same direction; because trailing teams also play eco rounds more often (12.3% of their CT rounds vs. 4.8% when leading), these two associations are not independent. In practice, the team fingerprints (Fig. 6(a)) and the similarity matrix (Fig. 6(b)) can support pre-match scouting: for example, opponents of NRG or Vitality can prepare T-side plans against C4, which each of these teams uses in 42% of its CT rounds. This extends the player-level modeling of Drachen et al. [25] to the level of team tactics.

C. Research Generalizability and Future Work

TAGNet's core framework, multi-snapshot temporal graph encoding with unsupervised clustering, is not specific to CS2. It could be combined with the VALORANT feature extraction of Hayakawa et al. [3] toward a unified FPS tactical analysis pipeline, and it brings graph representations of player interactions, as used by Raabe et al. [14] in invasion games, to the FPS domain. The approach may also transfer beyond games. DeepTrace [38] uses a GNN that updates its estimates as an epidemic contact network is explored during contact tracing, which parallels TAGNet's use of graph learning on evolving player interactions. An order-aware recurrent aggregator such as TAGNet's (Section III-D-2) could similarly summarize the order of recent contacts in contact-tracing graphs; we leave this adaptation to future work. Future work targets three directions: (a) *Multi-map generalization*: extending to Inferno and Mirage with cross-map transfer learning; (b) *Event-driven context modeling*: extending the per-snapshot context vector with intra-round events such as first-kill timing and utility

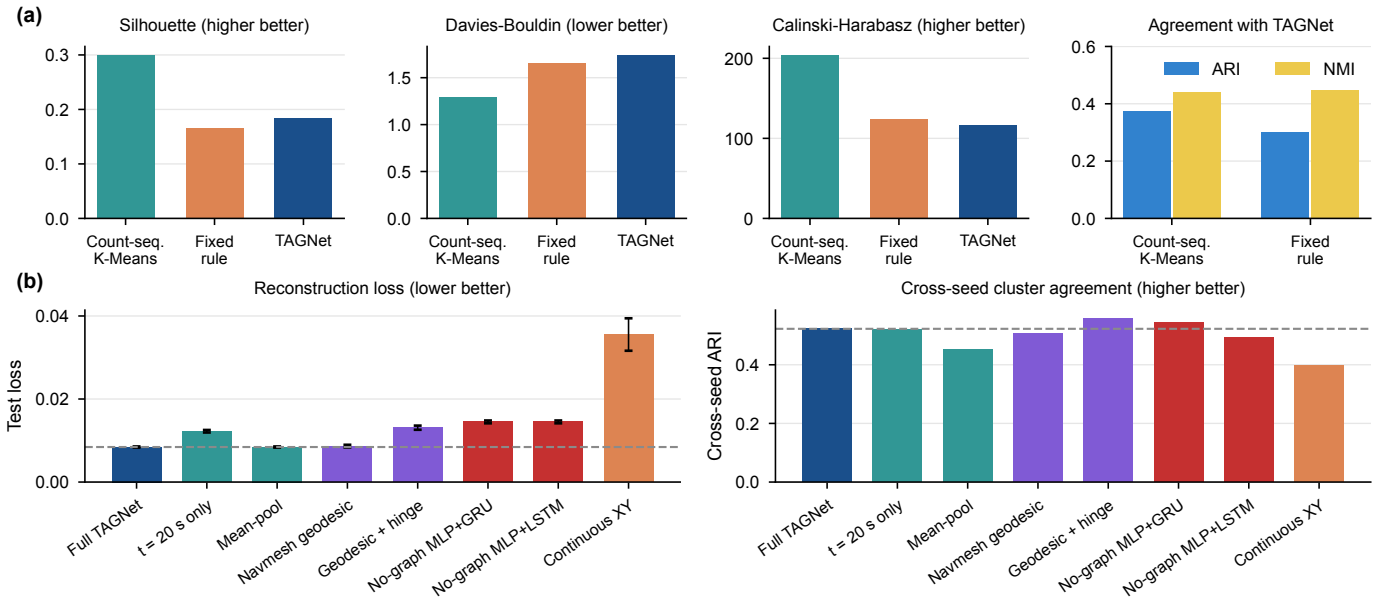


Fig. 9. Supplementary robustness checks. (a) Domain rule-based positional baseline vs. TAGNet: internal clustering metrics (SC, DBI, CH) and agreement (ARI, NMI) with TAGNet's K=6 partition. (b) Test loss (mean ± std over five seeds) and cross-seed ARI of the variants in Table VIII, relative to the full TAGNet configuration (dashed lines).

triggers; (c) *Real-time deployment*: integrating TAGNet's inference module into live broadcast systems for low-latency tactical recognition. Longer term, incorporating multimodal signals (field-of-view data, voice commands, and observer cameras) could move FPS tactical analysis from formation recognition toward decision prediction.

Beyond formation recognition itself, the six discovered patterns (C0–C5) and the round embedding $\mathbf{z}$ they are derived from have several potential downstream uses. For *opponent modeling*, TAGNet could be extended to predict, from a partially observed window, which of C0–C5 an opponent is converging toward, so that a counter-strategy can be chosen before the defensive setup is fully committed. For *game AI agent training*, C0–C5 provide a taxonomy of behavior targets derived from human play. This complements Nemobot Games [39], which uses large language models to build strategic game-playing agents for interactive learning. Nemobot focuses on *generating* agents and TAGNet on *analyzing* human tactics, so the patterns mined from professional play could serve as training targets for such agents, e.g., conditioning an agent to "play an anchor defense" (C0), with the context vector $\mathbf{c}_k$ (economy, equipment, score state) informing the choice of pattern. For *game design*, the taxonomy offers a basis for balance auditing, i.e., checking that each defensive pattern has a viable T-side counter-strategy. The round embedding $\mathbf{z}$ could also serve as a feature for round-outcome prediction, linking this work to the outcome-prediction literature of Section II. Finally, the fixed five-snapshot schedule is a design simplification rather than a claim that tactical timing is uniform; triggering snapshots on intra-round events (e.g., bomb plant, first contact) is a natural extension once event-timing annotations are available. Likewise, the single-map, CT-side scope is a choice for this first study rather than an architectural limit; T-side and multi-map extensions can reuse the same pipeline, with the map-specific changes noted in Section V-A.

VI. Conclusion

This paper proposes TAGNet for CT-side defensive tactical pattern recognition in professional Counter-Strike 2. TAGNet represents the five CT players at five snapshots per round as fully connected graphs, encodes each snapshot with a distance-aware graph attention encoder trained by self-supervised reconstruction, and aggregates the snapshots and per-snapshot context vectors in temporal order into round embeddings for K-Means clustering. On 697 CT defensive rounds, the final configuration lowers test-set reconstruction loss by 8.5% relative to an earlier configuration, multi-metric evaluation selects six interpretable defensive patterns (C0–C5), CT economy type is the contextual factor most strongly associated with tactical selection (χ^2=96.25, p<0.001), and aggressive forward pressure is used significantly more often when trailing (p<0.01). To our knowledge, this is the first unsupervised temporal graph learning framework for CT-side defensive tactical analysis in professional CS2 on Dust2; the pipeline and dataset construction can be extended to other maps, the T-side, and other multi-agent FPS titles. The released dataset, models, and code provide a reproducible starting point for such work.

Data Availability

The processed dataset of 697 CT-side rounds (player snapshots, pairwise distances, and per-snapshot match context), the trained models, and the code for demo parsing, dataset construction, training, and analysis are available at https://github.com/zhangx1ang03620/TAGNet-CS2 (mirror: https://github.com/weyoung-open-lab/TAGNet-CS2). The data were

derived from public HLTV.org match demos, which are listed in the repository.

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